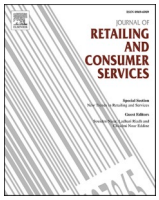
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The way out – Customer benefits and self-service satisfaction in cashierless shopping systems

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ABSTRACT

Introducing cashierless shopping systems offers an opportunity to innovate the checkout process for stationary retailers who encounter significant competition from both, existing stationary competitors and online retailers. The present study analyzes three types of cashierless shopping systems: mobile apps, face recognition, and self-checkouts. Following a review of customer benefits of digitization in stationary retailing, we integrate customer convenience, empowerment, and experience as well as self-service satisfaction in the Unified Theory of Acceptance and Use of Technology. An initial study with 406 participants validates the underlying framework while extending it with technology affinity and privacy concerns. Customer benefits in cashierless systems are explored based on 539 questionnaires. Performance expectancy, effort expectancy, customer empowerment, and customer experience positively impact shoppers' self-service satisfaction and willingness to use cashierless systems. Conversely, concerns about data privacy have a negative effect. The results reveal that participants perceive all types of cashierless shopping systems as comparatively similar. These systems perform the checkout process, empower shoppers, create positive service experiences, and adhere to data protection regulations. Qualitative insights ultimatively identify five key topics: convenience, security concerns, social concerns, technical concerns, and the shopping situation. These findings highlight the complexity of customer preferences regarding cashierless shopping systems.

1. Introduction

Stationary retail is highly competitive and faces numerous challenges. Competitors are within walking distance in city centers and malls (e.g., Fox and Sethuraman, 2006). Competitive pressure has been further increased by the establishment of retail models such as supercenters and hard discount retailers (e.g., Cleeren et al., 2009; Seiders et al., 2000). Manufacturers also intensify competition by engaging in fierce price negotiations against already low profit margins and disruption in the global supply chain (e.g., Gupta et al., 2021). At the same time, consumer behavior evolves with increasing expectations regarding omnichannel retailing and enhanced customer experience (Daugherty et al., 2019; Gauri et al., 2021).

The digital transformation also increases retail competition. Beyond electronic commerce, emerging business models, such as direct-to-consumer (e.g., McKee et al., 2023) and subscription models (e.g., Belavina et al., 2016; Wagner et al., 2021), transform the retail landscape. New digital competitors further intensify the competitive pressure. DoorDash, GoPuff, and Wolt, for example, offer last mile delivery in grocery retailing. Revenues of this online food delivery market (including meal delivery) are expected to grow to 1.86 trillion US dollars by 2029, including 1.35 trillion in grocery delivery (Statista Digital Market Insights, 2024).

Consequently, retailers need to differentiate from digital competition by creating compelling reasons for visiting physical stores (Breugelmans et al., 2023) and enhancing the overall shopping experience (Renko and Dreuzijanic, 2014; Schultz and Zacheus, 2024). Therefore, retailers prioritize experiential value, convenience, and personalized services, seeking to build emotional connections with customers, encourage repeat visits, and ultimately drive customer loyalty (Grewal et al., 2020; Reinartz et al., 2019).

The development of digital technologies also introduces innovations that retailers can employ to maintain competitiveness against e-commerce and digital challengers (Knof et al., 2024). Such emerging

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technologies can create value for stationary retailers by driving revenue growth through attracting new shoppers, increasing purchase volume among existing customers, or enhancing cost efficiency through self-service customers (Inman and Nikolova, 2017). Cashierless shopping systems may innovate the shopping experience by streamlining the checkout process.

The checkout process represents a critical service function in stationary retail that involves costly and time-intensive procedures (Pantano et al., 2023; Ponte and Bonazzi, 2023). Cashierless shopping systems may resolve these issues and enhance the shopping experience. They are related to fixed self-checkout systems in which customers use specific scanning devices at a fixed cash station (Andriulo et al., 2015). Two notable checkout systems are mobile apps and face recognition. With the mobile app, shoppers identify themselves upon entering the store (Cui et al., 2022) and the shopping basket is tracked through sensors, cameras, and artificial intelligence systems (Jordan et al., 2023). The shopper account is automatically charged when the customer exists the store. Amazon Go exemplifies this seamless "pick-and-go" shopping concept (Shankar et al., 2021).

An alternative approach involves payment through facial recognition technology (Gao et al., 2023; Lee and Pan, 2023). In this cashierless system, shoppers register an account with a service provider, such as AliPay or Tencent, linking their biometric data. Facial recognition is employed for customer identification while shopping and validation during the checkout (Pantano et al., 2023; Yu et al., 2024).

The present study aims to compare these two AI-driven processes with the comparatively established (traditional) self-checkout. For this purpose, we analyze customer empowerment, convenience, and experience as key customer benefits of digitization in stationary retailing (Reinartz et al., 2019) as part of shoppers technology acceptance (Venkatesh et al., 2003). The research model builds on the well-established Unified Theory of Acceptance and Use of Technology (UTAUT) model (Venkatesh et al., 2003) and integrates the relevant customer benefits (i.e., customer convenience, empowerment, and experience) to study innovative self-checkout systems. Whereas previous studies have focused on the rational perspective of using cashierless systems, the present study looks at broad customer benefits in conjunction with cognitive and normative values. Subero-Navarro et al. (2021), who use the cognitive-affective-normative (CAN) model, provide a comparative approach in the context of service robots. This study proposes the customer benefits adoption model which integrates the technological and customer value perspectives. The integrated research model (see Fig. 2) then answers the question of which factors determine shoppers' self-service satisfaction and willingness to use these self-service checkout systems.

2. Theoretical background

2.1. Cashierless shopping systems

Digital retail innovations particularly address customer experience to provide reasons to visit stationary retail outlets (Breugelmans et al., 2023). Self-checkout technologies (Andriulo et al., 2015) have already been installed to create value (Inman and Nikolova, 2017). This technological development has been particularly embraced by Amazon Go stores (Ives et al., 2019; Wankede et al., 2018). The innovative cashierless shopping systems generally include a cashless checkout by app authentication (Shankar et al., 2021) or biometrical authentication, such as face recognition (Lee and Pan, 2023). We briefly review the extant literature next. Schögel and Lienhard (2020) review key design features of the in-store purchase process in more detail.

Inman and Nikolova (2017) find general positive attitudes toward self-checkout, mobile apps, and scan and go approaches. These cashierless shopping systems are perceived with low to medium privacy concerns. Use intention of most systems is driven by performance expectancy and perceived risks (Ponte and Bonazzi, 2021).

Advantages of self-checkout systems are less people, no queues, and easy product organization, while their disadvantages are insufficient information, nonavailability of staff, and product scan difficulties (Knezevic et al., 2020). Self-efficacy, compatibility, and knowledge increase the acceptance of self-checkout systems through perceived usefulness and ease of use (Cebeci et al., 2020). Checkout quality, satisfaction, and loyalty is more relevant in staff checkouts compared to self-checkouts (Sharma et al., 2021).

High-convenience consumers favor walk-through checkouts to traditional cashiers, whereas low-convenience consumers show no preference (Ghazwani et al., 2022). App checkouts can also occasionally increase shopping convenience compared to self-service checkouts (van Esch et al., 2021). Due to by innovativeness importance, app checkouts increase level of arousal, perceived store atmosphere, and purchase intention (Cui et al., 2022). The intention to use app self-checkouts is positively driven by facilitating conditions, social influence, and openness to experience (Liang et al., 2022). Wankhede et al. (2018) exemplify some relevant situational challenges with Amazon Go. Chuawatcharin and Gerdsri (2019) also suggest that habitual behavior and smart card familiarity influence the adoption of a walk-through store. A text analysis of YouTube comments reveals that walk-through checkouts increase customers' shopping experience in the store, whereas concerns voiced are related to payments and job replacements of the service employees (Agarwal and Mittal, 2024).

Facial recognition is another innovative approach to checkout authentication. A recent literature review confirms that usage and user characteristics, along with privacy and security factors, influence the acceptance or resistance of facial authentication (Yu et al., 2024). These effects may be mediated by customer attitudes and evaluations. For example, customers' facial checkout acceptance is increased by perceived usefulness, security, visibility, expected effort, and social image, partly moderated by customer openness (Zhang and Kang, 2019). Similarly, perceived usefulness, ease of use, and perceived innovativeness positively affect customers' attitudes toward face checkouts, whereas perceived risks have a negative effect. Also, hedonic shopping value mediates the relationship between attitudes and store satisfaction (Dang et al., 2022). Customers would rather use facial checkout in stores than online (Moriuchi, 2021). The mediating role of trust and attitude are also stronger online than in stationary facial authentication. Customers experience a social presence effect (i.e., feeling watched by others) as well as a herding effect (i.e., following other customers). The social presence effect is mitigated by use experience (Gao et al., 2023).

Table 1 summarizes the findings of relevant studies regarding self-checkout, app, and face authentication in retailing.

2.2. Unified Theory of Acceptance and Use of Technology

The UTAUT proposes a unified view on competing models of technology acceptance (Venkatesh et al., 2003). After considerable utilization, this unified theory serves as a robust framework integrating new exogenous, endogenous, moderation, and outcome variables due to its comprehensive integration of various theoretical perspectives (Venkatesh et al., 2016). Regarding retail technologies, this unified theory has been applied in location-based services (Zhou, 2012), mobile banking (Jadil et al., 2021), self-service technologies (Blut et al., 2016), and voice assistants (Aw et al., 2022).

In the unified theory, four relevant variables (performance expectancy, effort expectancy, social influence, and facilitating conditions) have emerged that influence behavioral intention and, subsequently, actual use (Venkatesh et al., 2003): While performance expectancy represents the extent to which using the technology increases personal job performance, effort expectancy reflects users' perception of the technology's ease of use (Venkatesh et al., 2003). Social influence refers to "the degree to which an individual perceives that important others believe he or she should use the new system" (Venkatesh et al., 2003, 451). Facilitating conditions are the resources and support that allow the

Table 1
Overview of studies of cashierless systems.

Study	Cashierless System			Country	Industry	Study Type	N	Findings
	SC	AA	FR					
Wankhede et al. 2018		X		USA	Convenience	Case Study	–	Present the walk-through process and identify key challenges for the technology.
Chuawatcharin and Gerdsri (2019)		X		Thailand	Convenience	Survey	400	Validate the technology acceptance model without support for perceived entertainment, trust, and technology anxiety. Differences exist in demographics and shopping habits.
Zhang and Kang (2019)			X	China	not specified	Survey	299	Customers' facial checkout acceptance is increased by perceived usefulness, security, visibility, expected effort, and social image, partly moderated by customer openness.
Knezevic et al. (2020)	X			Croatia	Grocery	Survey	150	The advantages of the self-checkout system are no queues and easy product organization, while the disadvantages are insufficient information, nonavailability of staff, and scan difficulties.
Schögel and Lienhard (2020)	X	X	X	–	Electronics, Grocery, Convenience, Merchandise	Conceptual	12 retailer	The authors derive key design dimensions across the in-store purchase phases from the existing cashierless system.
Cebeci et al. (2020)	X			Turkey	Grocery	Survey	281	The authors test antecedents of technology acceptance in self-checkout systems and find positive effects of technology on self-efficacy, compatibility, and knowledge.
Moriuchi (2021)			X	USA	not specified	Survey	202 & 404	Customers prefer facial checkout in stores over online settings. Trust and attitude play a stronger mediating role online than in stationary retail. Self-efficacy moderates the relationships between performance expectancy, perceived risk, and customers' intention of use.
Ponte and Bonazzi (2021)	X	X		Italy	Grocery	Survey	789	Performance expectancy and privacy risk are key in all systems. Effort expectancy and facilitating conditions are not significant in walk-through technology.
Sharma et al. (2021)	X			UK & Austria	Grocery	Survey	313 & 474	Checkout quality, satisfaction, and loyalty are more relevant in staff checkouts compared to self-checkouts.
van Esch et al. (2021)	X	X		USA	not specified	Experiment	222, 509, 368	Compared to self-service checkouts, artificial intelligence (AI-)enabled checkouts creates perceived shopping convenience at times.
Cui et al. (2022)	X	X		USA	not specified	Experiment	271, 460, 660	Moderated by innovativeness importance, AI-enabled checkouts increase arousal, perceived store atmosphere, and purchase intent.
Dang et al. (2022)			X	China	Convenience, Department, Grocery	Survey	795	Perceived usefulness, ease of use, and perceived innovativeness positively affect customers' attitudes toward face checkouts, whereas perceived risks have a negative effect. Further, hedonic shopping value mediates the positive impact of attitudes on store satisfaction.
Ghazwani et al. (2022)		X		Saudi Arabia	Grocery	Online Experiment	328	High-convenience consumers favor walk-through checkouts over traditional checkouts. In contrast, low-convenience consumers show no preference.
Liang et al. (2022)		X		USA	Fashion	Survey	229	Facilitating conditions, social influence, and openness to experience increase intention to use app self-checkout.
Yong et al. (2022)	*	X		–	Grocery	Design	3 systems	Propose a prototype for a retail store on a small scale.
Gao et al. (2023)			X	China	Convenience	Econometric Model	134,924 transactions	Customers experience a social presence as well as a herding effect. The social presence effect is mitigated by use experience.
Jordan et al. (2023)		X		Germany	Grocery	Survey	302	Experiencing convenience at the checkout reduces customers' financial anxiety and leads to higher purchase intention.
Lee and Pan (2023)			X	China	not specified	Survey	321	Attractiveness and security of the checkout process increase perceived performance and effort expectancy and create positive emotions leading to continued use intention.
Pantano et al. (2023)			X	Italy	not specified	Interview	80	Neither age nor technology knowledge impact the perception of locus of control over personal data collection. However, expected economic and utilitarian gratifications differ across age groups.
Agarwal and Mittal 2024		X		–	Convenience	Text analysis	7348 comments	Walk-through checkouts increase customers' shopping experience. Customers' concerns are related to payments and job replacements.
Yu et al. (2024)			X	–	–	Review	22 studies	Identify 37 factors (related to use, user, attitude & evaluation, privacy & security, and others) that influence acceptance or resistance of facial recognition in retail.
This Study	X	X	X	Germany	Grocery	Survey, Interview	406, 539, 124	Customers perceive the types of cashierless shopping systems as relatively similar. These systems perform the checkout, empower shoppers, create positive service experiences, and adhere to data protection. Qualitative findings reveal five primary topics: convenience, security

(*continued on next page*)

Table 1 (*continued*)

Study	Cashierless System			Country	Industry	Study Type	N	Findings
	SC	AA	FR					
								concerns, social concerns, technical concerns, and the shopping situation.

Notes: SC: Self-checkout, AA: App authentication, FR: Facial recognition. *: Regi-Robo and Uniqlo system.

effective use of the technology. The approach also controls for interactions related to age, gender, experience, and voluntariness of use.

The cognitive variables (performance expectancy and effort expectancy) also represent one part of the cognitive-affective-normative (CAN) model. Social influence may also reflect the normative part (Subero-Navarro et al., 2021). Subero-Navarro et al. (2021) use the CAN model to reflect on technophobia when engaging with a service robot in-store. Following the significant negative relationship between technophobia and intention to use service robots, this study considers technology affinity as a positive influence on this relationship. In their study, the affective dimension is represented by arousal and pleasure – elements that are related to customer benefits identified by Reinartz et al. (2019) (i.e., customer convenience and experience).

2.3. *Customer benefits of digitization in stationary retailing*

Retailers can add value when they satisfy their customers' needs (Reinartz et al., 2019; Shankar et al., 2021). Consumer perceived value often originates in the discussion of quality and price. For example, perceived value is defined as a "consumer's overall assessment of the utility of a product (or service) based on perceptions of what is received and what is given." (Zeithaml, 1988, 14) Zeithaml acknowledges different and differential effects of quality and price on perceived value. Research has subsequently argued for a more complex view on consumer perceived value (e.g., Walsh et al., 2014).

The digital transformation innovates value creation to meet customer needs (Reinartz et al., 2019). The retail value chain for digital technologies is impacted by five key customer benefits: convenience, empowerment, experience, relevance, and savings (Reinartz et al., 2019). We briefly discuss these five customer benefits in the context of cashierless systems.

Online retailers are said to provide more convenience than stationary retailing due to anywhere and anytime shopping convenience (Duarte et al., 2018). Convenience generally encompasses all factors that stimulate physical or mental ease and simplify the fulfillment of customer needs. (Reinartz et al., 2019). Consequently, customers' shopping convenience encompasses several dimensions (Duarte et al., 2018). With regard to cashierless systems, customer convenience is primarily based on saving time (Duarte et al., 2022; Schultz and Zacheus, 2024).

Customer empowerment refers to "a positive subjective state evoked by consumer perceptions of increasing control" (Hunter and Garnefeld, 2008, 1). Customers perceive to be in control of self-service technologies, such as cashierless systems, when their design is not too complex (Zhu et al., 2007). Customers value systems that are easy to understand and use more – moderated by prior experience and technology readiness (Zhu et al., 2007). More generally, Reinartz et al. (2019) propose that empowerment increases customers' control over their actions, thus improving the customer decision process.

Creating superior customer experience is proposed as one central objective in retail environments (Verhoef et al., 2009), that are constructed over multiple touchpoints (Lemon and Verhoef, 2016) and may include cognitive, affective, social, and physical aspects (Verhoef et al., 2009). Integrating cashierless checkout processes into customers' shopping routines enables these systems to establish a presence within the immediate retail environment (Reinartz et al., 2019), potentially facilitating a seamless incorporation into daily life if customers engage with them (Gong et al., 2022).

Customer relevance refers to the degree to which information is valuable and beneficial for customers (Reinartz et al., 2019). Following the UTAUT (Venkatesh et al., 2003, 2016), relevance is primarily attributed through performance and effort expectancy of cashierless systems.

Lastly, monetary savings are generated on the firm level through optimized product and resource usage (Reinartz et al., 2019). On the customer level, customers could receive direct monetary benefits by using cashierless systems. While such approaches are conceivable, they have not yet been implemented in current cashierless systems.

3. Research process and research model

The present study employs a multi-step research process to address the research questions. First, the research answers the general question of whether customers are willing to adopt and use cashierless shopping systems. The first study builds on the well-established UTAUT (Venkatesh et al., 2003), additionally including technology affinity as one relevant driver (e.g., Dang et al., 2022) and data privacy as one relevant barrier (e.g., de Bellis and Johar, 2020; Ponte and Bonazzi, 2021) of cashierless shopping systems. Thus, the primary aim of the first study is theory confirmation.

The second study extends the first one, and its aim is theory development. Following key customer benefits of digitization in stationary retailing (Reinartz et al., 2019), Study 2 integrates the central benefits (i.e., convenience, empowerment, experience, and relevance) in the research model. Additionally, the study considers self-service satisfaction (e.g., Sharma et al., 2021) as another direct outcome of the cashierless system encounter. Self-service satisfaction mediates the effect of the proposed antecedents on customers' intention to use the systems. Study 2 answers the research question of which customer benefits contribute to the satisfaction and intended use of cashierless shopping systems.

Finally, Study 3 uses a qualitative approach and focusses on theory exploration. The study purpose is to identify further relevant factors perceived by customers, extending our understanding of their willingness or resistance to engage with cashierless shopping systems.

Fig. 1 summarizes and outlines the research process.

Fig. 2 displays the research model. It is further discussed in section 4. The research model builds on the factors that influence consumers' satisfaction with and their intention to use cashierless systems. The factors are adopted from relevant literature decribed in section 2.

In detail, we propose a customer benefit adoption model that augments the UTAUT model by the customer value perspective. Digital innovations are said to create customer value by convenience, empowerment, and experience. While these customer benefits are considered in the extant literature, they have not been included in the UTAUT, which primarily builds on the rational perspective of the technology (i.e., performance and effort expectancy). The proposed customer benefit adoption model includes the interrelationship between both perspectives and relevant outcomes in self-service satisfaction and customer behavioral intention. In addition to that scientific novelty, we further control for demographics and types of cashierless systems. Altogehter, the research model helps to better understand the dynamics in creating customer satisfaction as well as customers' intention to use cashierless systems.

	Theory Purpose	Research Type	Research Question	n
Study 1	Theory Confirmation	Quantitative	RQ1: Are customers willing to adopt and use cashierless shopping systems?	406
Study 2	Theory Development	Quantitative	RQ2: Which customer benefits contribute to the satisfaction and use of cashierless shopping systems?	539
Study 3	Theory Exploration	Qualitative	RQ3: What are the customer-relevant factors that contribute to the understanding of cashierless shopping systems?	124

Fig. 1. Overview of the research process.

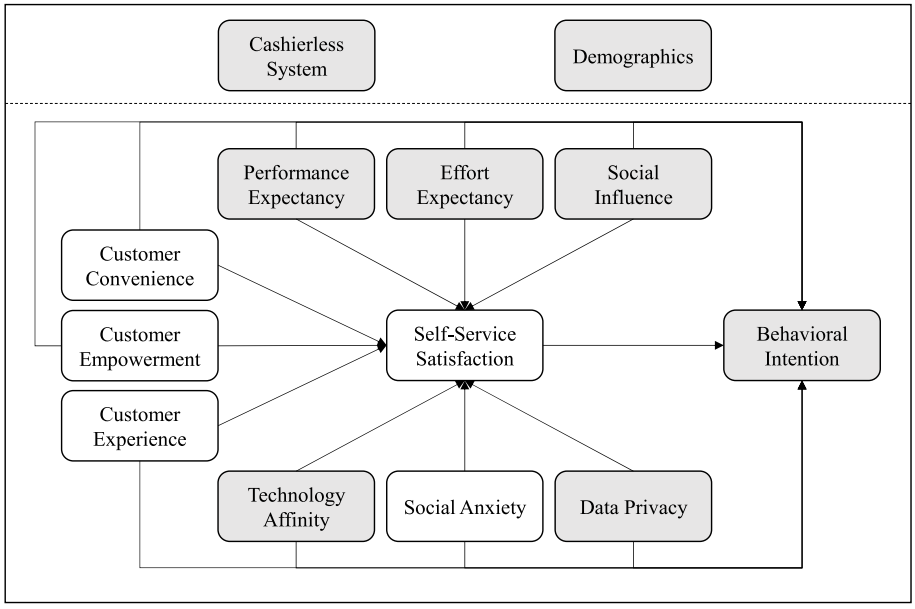


Fig. 2. Research model for the quantitative studies 1 and 2.
Notes: Shaded variables are included in studies 1 and 2.

4. Preliminary study on the acceptance of cashierless systems (study 1)

4.1. Design of study 1

A self-report survey was conducted to access customers' general willingness to accept and use cashierless systems (see shaded areas in Fig. 2). The questionnaire asked for participants' signed consent to anonymously and voluntarily participate in the research survey. Eleven participants pretested the questionnaire to ensure its functionality and comprehensibility. The questionnaire introduces cashierless systems. It first establishes participants' disposition regarding technology affinity. Next, participants are randomly assigned to the innovative types of cashierless shopping, mobile apps and face recognition. Following a between-subject design, each participant only sees one type of cashierless shopping. Both types are subsequently presented in text and pictures. Participants then answer questions about perceived data privacy (Inman and Nikolova, 2017) and on the UTAUT (Venkatesh et al., 2003). At the end, the questionnaire collects demographic details.

All measurements are derived from the extant literature and adapted to the context of cashierless shopping systems. The items are measured by 7-point rating scales, ranging from 1 ("strongly disagree") to 7 ("strongly agree"), and 7-point semantic differentials. Following the framework (Venkatesh et al., 2003), we measure performance expectancy, effort expectancy, social influence, and intention to use. We also capture age, gender, and experience. Additionally, the research study controls for the type of cashierless system. 5- and 4-item scales are used for performance and effort expectancy, respectively (Im et al., 2011; Zhou, 2012). Social influence is measured by 4 items by Walsh et al. (2013). The outcome variable is 'intention to use cashierless systems' (9 items by Brooke, 1996). Study 1 also registers participants' technology affinity (9 items by Franke et al. (2019)) and data privacy (3 items by Zhang et al., 2019) as additional variables.

4.2. Data collection and sample

Data collection ran in January and February 2022. We utilized a convenience sample to efficiently gather insights from participants who were readily accessible and met the study's general criteria. The online survey provided a final sample of 406 participants. Following the

between-subject design, 209 were confronted with the mobile app and 197 were asked about the face recognition process. The final sample thus exceeded the minimum sample size of n = 138 (effect size = 0.15; α error = 0.05; power 0.95).

229 participants identified as female (56.4%), 177 as male (43.6%), and no participants identified as diverse. Mean age was 33.9 years (SD = 12.4). 129 (31.8%) participants declared previous experience with cashierless systems. Study 1 sample characteristics are available in the appendix.

4.3. Data analysis and results

We conducted covariance-based structural equation modeling to analyze the relationships in the research model (Fig. 2). Data analysis was performed with the R package lavaan (Rosseel, 2012). All measurement models were satisfactory regarding individual item reliability, construct reliability, and discriminant validity. The global fit indexes also propose an overall good fit ($\chi 2 = 998.61$, df = 329, χ 2/df = 3.035, CFI = 0.928, TLI = 0.917, RMSEA = 0.071, SRMR = 0.048), supporting the reliability and validity of the measurement models.

For the structural model, the global fit indexes also support a good fit to the data ($\chi 2 = 1422.95$, df = 672, χ 2/df = 2.117, CFI = 0.921, TLI = 0.900, RMSEA = 0.074, SRMR = 0.051). Both, customers' performance expectancy ($\beta_{app} = 0.198$, $p = 0.002$; $\beta_{face} = 0.466$, $p < 0.001$) and social influence ($\beta_{app} = 0.498$, $p = 0.002$; $\beta_{face} = 0.466$, $p < 0.001$) positively impact customers' willingness to use the innovative cashierless systems. Effort expectancy significantly affects customers' willingness to use in the app case but not in the face authentication scenario ($\beta_{app} = 0.174$, $p = 0.048$; $\beta_{face} = 0.079$, $p = 0.375$). The empirical results further support the negative impact of perceived data privacy on customers' intention to use these systems ($\beta_{app} = -0.193$, $p = 0.003$; $\beta_{face} = -0.319$, $p < 0.001$). Customers' technology affinity do not impact their behavioral intention ($\beta_{app} = 0.121$, $p = 0.103$; $\beta_{face} = 0.033$, $p = 0.660$). For the demographic controls, the results was not statistically different for younger (<30) and older (≥30) participants. Previous experience primarily interacts with technology affinity but is not statistically significant in both groups. The results show two relevant interaction effects; (a) gender and performance expectancy ($\beta_{app} = -0.121$, $p = 0.038$; $\beta_{face} = 0.140$, $p = 0.022$) and (b) gender and social influence ($\beta_{app} = -0.104$, $p = 0.025$; $\beta_{face} = -0.117$, $p = 0.088$). So, the effect of performance expectancy on behavioral intention is higher for men, and the relationship between social influence and behavioral intention is more pronounced for women. Table 2 summarizes the structural model results.

Table 2
Multi-group analysis of the structural model for app and face authentication.

Independent Variable	Dependent Variable	App Authentication		Face Authentication	
		Std. Path. Estimate	*p*-value	Std. Path. Estimate	*p*-value
Performance Expectancy	Behavioral Intention	0.498	0.002	0.466	0.000
Effort Expectancy	Behavioral Intention	0.174	0.048	0.079	0.375
Social Influence	Behavioral Intention	0.198	0.000	0.180	0.013
Technical Affinity	Behavioral Intention	0.121	0.103	0.033	0.660
Data Privacy	Behavioral Intention	−0.193	0.003	−0.319	0.000
Age	Behavioral Intention	0.066	0.148	0.088	0.064
Gender x Performance Expectectancy	Behavioral Intention	0.121	0.038	0.140	0.022
Gender x Social Influence	Behavioral Intention	−0.104	0.025	−0.117	0.088
Experience x Technical Affinity	Behavioral Intention	0.031	0.500	0.076	0.140

5. Study 2

5.1. Design of study 2

In Study 2, we analyze the role of customer benefits in cashierless shopping systems through an online survey. Before randomly assigning participants to the three types of cashierless shopping systems, we establish participants' general disposition regarding technology affinity and social anxiety. Again, each participant only sees one type of cashierless shopping, following a between-subject design. The cashierless shopping instances of mobile apps, face recognition, and self-checkout are explained in text and supported by videos. Afterwards, we ask about the explanatory and explained variables. At the end of the questionnaire, we gather demographic details for control.

Study 2 follows the outline of Study 1. Adding to participants' disposition regarding technology affinity and data privacy (Study 1), Study 2 also captures participants' social anxiety by a 6-item scale (Fenigstein et al., 1975).

Study 2 extends Study 1 further by measuringcustomer benefits following Reinartz et al. (2019). As there are generally no direct monetary savings by cashierless systems and their relevance is already covered by performance and effort expectancy, we concentrate on customer convenience (6 items by Duarte et al., 2022), empowerment (5 items based on Hunter and Garnefeld, 2008; Zhu et al., 2007), and experience (4 items by Gong et al., 2021). As outcome variables, self-service satisfaction is ascertained with a 4-item scale (Sharma et al., 2021) and intention to use cashierless systems adapted from the 9-item measurement (Brooke, 1996). Again, all measurement items are derived from the extant literature, adapted to the context of cashierless shopping systems, and measured by 7-point rating scales, ranging from 1 ("strongly disagree") to 7 ("strongly agree"), and 7-point semantic differentials.

5.2. Data collection and sample

We conducted an online survey through a research panel over 2 months at the beginning of 2024 including one reminder message. The likelihood of overlap between the sample of Study 1 and Study 2 is marginally small due to the different sampling approaches. Beyond the established scale measures, we also collected demographic data as control variables. All questionnaires were checked for completeness, response behavior ("straightlining"), and processing time (relative speed index RSI ≤2; Leiner, 2019). After data cleaning, the final sample included 539 questionnaires, thus exceeding the minimum sample size of n = 160 (effect size = 0.15; α error = 0.05; power 0.95). In the final sample, 179 completed the questionnaire for the traditional self-checkout, 177 for the mobile app variant, and 183 for the face recognition system.

The sample is characterized by 51.4% (277) female, 44.5% (240) male, 1.1% (6) diverse, and 3.0% (16) unknown. The mean age is 45.6 (sd = 6.06). Median net household income per month is 4000 to 4999 Euro, and the categorized mean value is 3582 Euro. Table 3 details the sample characteristics. The sample's demographic distributions mirror those in Germany: In 2023, 50.7% of people lving in Germany identified as female (Destatis, 2024), the mean age of German respondents was 44.6 years (Destatis, 2025), and the average monthly household income constituted 3650 Euro (Kott, 2024).

5.3. Data analysis and results

For Study 2, variance-based structural equation modeling is used and estimation is performed via the R package plspm (Sanchez et al., 2015). The variance-based approach has minimal distributional requirements and low computational demands for the underlying algorithm (Hair

Table 3
Overview of sample characteristics.

Variable		Absolute Frequency	Relative Frequency
Gender	female	277	51.4%
	male	250	44.5%
	diverse	6	1.1%
Age	≤19	10	1.9%
	20–29	53	9.8%
	30–39	116	21.5%
	40–49	127	23.6%
	50–59	137	25.4%
	60-69+	63	11.7%
	≥70	15	2.8%
Net Household Income	<1000	16	3.0%
	1000–1999	59	10.9%
	2000–2999	94	17.4%
	3000–3999	84	15.6%
	4000–4999	70	13.0%
	≥5000	113	21.0%
	w/o answer	103	19.1%

et al., 2020). It allows for complex models and facilitates the prediction and explanation of dependent variables within a theoretically grounded structural model (Hair et al., 2021; Sarstedt et al., 2014). Consequently, the variance-based structural equation model typically provides high statistical power for hypothesis testing for the hypothesis testing presented in the research model (see Fig. 2).

First, we assess individual item reliability, composite reliability, and discriminant validity of the measurement models, employing 5000 bootstrap resamples to obtain robust standard errors. Beyond the structural designs against common method bias, we also calculate Harman's single-factor test. A single factor explains 32.2% of the variance, thus remaining below the threshold of 0.50. We also control for response bias by comparing the first half to the second half of responses and find no significant differences.

The measurement models demonstrate reliable and valid measurement overall (see Tables 4 and 5). Considering indicator reliability, five measurement items must be excluded due to low loadings (TA6 = 0.563, BI3 = 0.433, BI6 = 0.408, BI7 = 0.578, BI9 = 0.467). All other average loadings of the measurement items lie between 0.632 and 0.974 and exceed the required value of 0.6 for good construct reliability. The confidence interval of social anxiety already shows considerable levels of variance. The first (greater 1; between 2.781 and 5.072) and second (below 1; between 0.149 and 0.787) eigenvalues also suggest a single construct solution per item measurement scale. We can also assume construct reliability with values for Cronbach's alpha (between 0.820 and 0.961) and Dillon-Goldstein's rho (between 0.875 and 0.974) over 0.7 for all latent constructs. All constructs also have an average variance extracted (AVE) of 0.50 (between 0.581 and 0.927), so convergence validity can be assumed.

The Fornell-Larcker criteria, cross-loadings, and the heterotrait-monotrait ratios 2 (HTMT2) support discriminant validity. All values of construct correlations are less than the square root of the AVE. Following the cross-loadings criteria, each item has the highest loading with its respective construct. The HTMT2 values range between 0.034 and 0.849, all below the conservative threshold value of 0.85. Table 6 reports the Fornell-Larcker criteria and the HTMT2 values.

The structural model yields mixed results regarding the proposed research model. Regarding the underlying framework, performance expectancy positively affects self-service satisfaction ($\beta = 0.201$, $p < 0.001$) and behavioral intention ($\beta = 0.130$, $p = 0.002$), whereas effort expectancy only increases behavioral intention significantly ($\beta = 0.160$, $p < 0.001$) but not satisfaction ($\beta = -0.007$, $p = 0.824$). Social influence has no significant effect on the outcome variables ($\beta = -0.008$, $p = 0.787$, $\beta = -0.035$, $p = 0.137$). The customer benefits proposed by Reinartz et al. (2019) reveal no significant impact of customer convenience ($\beta = -0.007$, $p = 0.847$, $\beta = 0.048$, $p = 0.111$), but empowerment

Table 4
Measurement results of the item loadings.

Latent Variable	Item	Loading	Std. Error	LLCI	ULCI
Performance Expectancy (PE)	PE1	0.906	0.010	0.885	0.924
	PE2	0.834	0.016	0.801	0.864
	PE3	0.875	0.012	0.850	0.897
	PE4	0.875	0.013	0.849	0.898
	PE5	0.897	0.011	0.874	0.917
Effort Expectancy (EE)	EE1	0.890	0.012	0.864	0.912
	EE2	0.843	0.020	0.802	0.879
	EE3	0.904	0.012	0.880	0.925
	EE4	0.896	0.015	0.862	0.922
Social Influence (SI)	SI1	0.878	0.017	0.841	0.909
	SI2	0.919	0.012	0.894	0.939
	SI3	0.875	0.013	0.848	0.900
	SI4	0.883	0.026	0.822	0.922
Technology Affinity (TA)	TA1	0.872	0.017	0.833	0.898
	TA2	0.882	0.013	0.854	0.904
	TA3	0.653	0.035	0.582	0.718
	TA4	0.825	0.020	0.779	0.857
	TA5	0.860	0.015	0.826	0.885
	TA6	*0.563*	—	—	—
	TA7	0.691	0.038	0.605	0.755
	TA8	0.685	0.036	0.608	0.746
	TA9	0.829	0.017	0.792	0.860
Data Privacy (DP)	DP1	0.948	0.007	0.933	0.962
	DP2	0.974	0.004	0.965	0.981
	DP3	0.967	0.005	0.956	0.975
Behavioral Intention	BI1	0.864	0.010	0.844	0.881
	BI2	0.632	0.033	0.564	0.694
	BI3	*0.433*	—	—	—
	BI4	0.798	0.019	0.757	0.830
	BI5	0.659	0.034	0.590	0.722
	BI6	*0.408*	—	—	—
	BI7	*0.578*	—	—	—
	BI8	0.830	0.015	0.799	0.860
	BI9	*0.467*	—	—	—
Customer Convenience (CC)	CC1	0.632	0.042	0.544	0.708
	CC2	0.850	0.018	0.811	0.882
	CC3	0.783	0.025	0.728	0.828
	CC4	0.858	0.014	0.828	0.884
	CC5	0.854	0.014	0.827	0.880
	CC6	0.837	0.023	0.788	0.877
Customer Empowerment (CE)	CE1	0.832	0.015	0.801	0.861
	CE2	0.886	0.009	0.867	0.904
	CE3	0.832	0.019	0.792	0.865
	CE4	0.744	0.026	0.689	0.793
	CE5	0.640	0.034	0.568	0.703
Customer Experience (CX)	CX1	0.942	0.004	0.933	0.950
	CX2	0.941	0.006	0.927	0.952
	CX3	0.794	0.023	0.745	0.835
	CX4	0.865	0.015	0.832	0.892
Social Anxiety (SA)	SA1	0.868	0.096	0.596	0.910
	SA2	0.678	0.126	0.373	0.844
	SA3	0.781	0.106	0.497	0.871
	SA4	0.806	0.108	0.507	0.904
	SA5	0.766	0.121	0.429	0.857
	SA6	0.853	0.111	0.521	0.909
Self-Service Satisfaction	SS1	0.927	0.006	0.915	0.937
	SS2	0.895	0.011	0.872	0.914
	SS3	0.922	0.009	0.902	0.938
	SS4	0.919	0.010	0.898	0.936

Notes: items in italics dropped; other values after item reduction, LLCI: lower level confidence interval, ULCI: upper level confidence interaval.

($\beta = 0.234$, $p < 0.001$) and experience ($\beta = 0.397$, $p < 0.001$) significantly increase self-service satisfaction. Empowerment ($\beta = 0.223$, $p < 0.001$) and experience ($\beta = 0.185$, $p < 0.001$) also positively impact behavioral intention. For the boundary conditions, technology affinity ($\beta = -0.052$, $p = 0.067$, $\beta = -0.007$, $p = 0.773$) and social anxiety ($\beta = 0.018$, $p = 0.496$, $\beta = 0.004$, $p = 0.843$) do not significantly impact self-service satisfaction and behavioral intention. Perceived data privacy reveals the expected negative effect on satisfaction ($\beta = -0.109$, $p < 0.001$) and behavioral intention ($\beta = -0.192$, $p < 0.001$). Finally, self-service satisfaction significantly increases shoppers' behavioral

Table 5
Measurement results per latent variable.

Latent Variable	Alpha	Rho	1st EV	2nd EV	AVE
Performance Expectancy (PE)	0.925	0.944	3.855	0.403	0.770
Effort Expectancy (EE)	0.906	0.934	3.124	0.382	0.781
Social Influence (SI)	0.913	0.939	3.172	0.386	0.790
Technology Affinity (TA)	0.915	0.932	5.072	0.787	0.627
Data Privacy (DP)	0.961	0.974	2.781	0.149	0.927
Behavioral Intention	0.820	0.875	2.921	0.764	0.581
Customer Convenience (CC)	0.891	0.918	3.909	0.676	0.650
Customer Empowerment (CE)	0.849	0.893	3.138	0.678	0.627
Customer Experience (CX)	0.910	0.937	3.155	0.452	0.788
Social Anxiety (SA)	0.884	0.912	3.813	0.614	0.631
Self-Service Satisfaction (SS)	0.936	0.954	3.355	0.317	0.838

Notes: Alpha: Cronbach's alpha; Rho: Dillon-Goldstein rho, EV: eigenvalue, AVE: average variance extracted.

intention ($\beta = 0.180$, $p < 0.001$). Table 7 summarizes the relational results of the structural model.

The explained variance of self-service satisfaction is satisfactory ($R^2 = 64.7\%$). Overall, the research model explains 77.2% in variance of shoppers' intention to use cashierless systems.

Results may also differ for socio-demographics. While participants' age and gender show no statistically significant differences, monthly net income reveals one significantly different relationship. Social influence hurts the behavioral intention of households with lower income but is insignificant for households with higher income ($\beta_l = -0.096$, $\beta_h = 0.003$, $p = 0.017$). However, the value is just close to the 0.1 relevance threshold.

The comparisons between the three cashierless shopping systems yield only minor differences. Comparing the traditional self-checkout with mobile app cashierless, technology affinity had a negative effect on satisfaction for the mobile app scenario ($\beta_s = 0.021$, $\beta_m = -0.133$, $p = 0.015$). Similarly, participants perceive convenience in the self-checkout scenario but not in face recognition ($\beta_s = 0.131$, $\beta_f = -0.025$, $p = 0.025$). Mobile apps and face recognition also show only one significantly different relationship. Social interaction on satisfaction is relevant for face recognition but not for the mobile app scenario ($\beta_f = 0.080$, $\beta_m = -0.078$, $p = 0.007$), but both coefficients are below the 0.1 relevance criteria.

6. Qualitative results (study 3)

Participants of Study 2 were invited to answer an open question regarding the cashierless shopping systems. This question constitutes Study 3. 124 participants (23.0% response rate) voluntarily commented on the cashierless systems. Respondents were 46.8% (n = 58) female, 50.0% (n = 62) male, and 3.2% (n = 4) diverse, and on average 48.9 (sd = 13.4) years old. Both authors reviewed all responses, identified key statements, and categorized the statements into dominant topics (initially independently and then combined). Five key topics emerged from the qualitative analysis: convenience, security concerns, social concerns, technical concerns, and shopping situation. Based on the categorization, unweighted Cohen's kappa results in 0.920. The lower level of the confidence interval is 0.867, whereas the upper level is 0.973. The kappa of 0.920 exceeds the 0.81 level suggested by Landis and Koch (1977) as representing almost perfect agreement.

Respondents generally attribute convenience to time-saving. This is consistent with existing results (e.g., Benoit et al., 2024; Schultz and Zacheus, 2024). However, they also indicate displeasure with these systems. For example, one respondent states, "I am much slower than a cashier." Similarly, "the checkout is the job of retailers not customers.", as one person argues, "If I do their job, I want a 2% discount." The responses also indicate that the introduction of cashierless systems does not remove waiting lines per se. Instead, waiting lines still exist with self-checkouts. People also fear that waiting may increase when more people, who are not willing or are unable to use the cashierless system, are in line for fewer cashiers. This would constitute another form of digital divide.

Respondents also tie convenience to the shopping situation. One respondent formulates, "With just a few items, I am certainly faster than at a normal checkout – but not with a full shopping cart where handling becomes more complicated." Beyond the number of items, respondents also consider the number of customers and the storage area at the cashierless checkout. Some also state that their responses depend on whether the shopping situation is relaxed or stressful. Consequently, one person states, "Since 2020, I've been using the Rewe pick-up service more and more to keep my shopping as short as possible."

Respondents also raise concerns against cashierless systems. Security concerns primarily focus on fear of total surveillance and loss of data privacy, resulting in doubts and anxieties about storage, use, and transfer of data: "This system opens the door to complete surveillance."

Regarding technical concerns, respondents are primarily concerned with system malfunctions. For example, "the existing self-checkout systems never work smoothly". "I have no problem with the technology, but I have no confidence in the reliability of the item recording." Examples of such concerns are the system identifying articles wrongly, determining a wrong article quantity, and charging the wrong prices. People are also concerned that the system might charge their account multiple times. With the process shifting from cashiers to customers, customers express concerns about being perceived as engaging in theft when using self-checkout systems. One participant notes, "It sometimes feels like you're stealing, which can create discomfort." Similarly, concerns arise when complaining, where complaints about missing items may lead to accusations of dishonesty. As one participant describes, "When you regularly report missing products, some employees assume you have taken the item yourself or given it to someone else, essentially accusing you of attempted theft."

The most pronounced concerns are social ones. Respondents fear

Table 6
Assessment of discriminant validity.

	PE	EE	SI	TA	DP	BI	CC	CE	CX	SA	SS
PE	**0.877**	0.505	0.325	0.290	−0.382	0.748	0.675	0.765	0.787	0.067	0.709
EE	0.550	**0.884**	0.119	0.312	−0.340	0.582	0.534	0.474	0.492	−0.086	0.415
SI	0.340	0.122	**0.889**	0.077	−0.178	0.297	0.229	0.412	0.403	0.094	0.328
TA	0.281	0.319	0.056	**0.792**	−0.127	0.273	0.201	0.247	0.361	0.024	0.217
DP	0.392	0.361	0.179	0.092	**0.963**	−0.590	−0.238	−0.524	−0.466	0.088	−0.480
BI	0.804	0.690	0.246	0.261	0.632	**0.762**	0.577	0.784	0.775	0.022	0.745
CC	0.712	0.590	0.208	0.174	0.244	0.639	**0.806**	0.567	0.560	0.008	0.493
CE	0.814	0.529	0.427	0.192	0.560	0.849	0.599	**0.792**	0.771	0.082	0.729
CX	0.843	0.542	0.425	0.374	0.454	0.781	0.590	0.800	**0.888**	0.066	0.758
SA	0.060	0.082	0.107	0.034	0.083	0.079	0.036	0.102	0.063	**0.794**	0.066
SS	0.743	0.442	0.341	0.204	0.497	0.761	0.508	0.774	0.787	0.061	**0.915**

Notes: PE: performance expectancy, EE: effort expectancy, SI: social influence, TA: technology affinity, DP: data privacy, BI: behavioral intention, CC: customer convenience, CE: customer empowerment, CX: customer experience, SA: social anxiety, SS: self-service satisfaction; bolded values represent square roots of the average variance extracted. Values in the lower triangle represent HTMT2, whereas the upper triangle reports correlations between constructs.

Table 7
Structural model results (study 2).

Independent Variable	Dependent Variable	Std. Path Estimate	*p*-value	Support
Performance Expectancy	Self-service Satisfaction	0.201	<0.001	Yes
Effort Expectancy	Self-service Satisfaction	−0.007	0.824	No
Social Influence	Self-service Satisfaction	−0.008	0.787	No
Technology Affinity	Self-service Satisfaction	−0.052	0.675	No
Social Anxiety	Self-service Satisfaction	0.018	0.496	No
Data Privacy	Self-service Satisfaction	−0.109	<0.001	Yes
Customer Convenience	Self-service Satisfaction	−0.007	0.847	No
Customer Empowerment	Self-service Satisfaction	0.234	<0.001	Yes
Customer Experience	Self-service Satisfaction	0.397	<0.001	Yes
Performance Expectancy	Behavioral Intention	0.130	0.002	Yes
Effort Expectancy	Behavioral Intention	0.160	<0.001	Yes
Social Influence	Behavioral Intention	−0.035	0.137	No
Technology Affinity	Behavioral Intention	−0.007	0.773	No
Social Anxiety	Behavioral Intention	0.004	0.843	No
Data Privacy	Behavioral Intention	−0.192	<0.001	Yes
Customer Convenience	Behavioral Intention	0.048	<0.001	Yes
Customer Empowerment	Behavioral Intention	0.223	0.111	No
Customer Experience	Behavioral Intention	0.185	<0.001	Yes
Self-service Satisfaction	Behavioral Intention	0.180	<0.001	Yes

digital divide for elderly or people with disabilities. They also miss social interactions or fear social isolation. “I’d rather have a good-humored person at the checkout who is also employed, instead of having to do everything myself.” Respondents envision job cuts in retailing as retailers aim at cutting staffing costs. This is generally in line with Sharma et al. (2021). One respondent frames cashierless systems as “Another step towards cutting staff and ‘dehumanizing’ the system.”

7. Discussion

Cashierless shopping systems provide opportunities for stationary retailers to increase the overall shopping experience and reduce shoppers’ waiting time for the checkout process. The empirical results consequently find that customer experience with these systems is relevant to shoppers’ self-service satisfaction and their willingness to use such systems. Agrawal and Mittal (2024) already posit that these systems create customer experience. Comparable in spirit of Jordan et al. (2023) showing that a convenient experience mitigates financial anxiety, our findings also support that perceived customer experience creates self-service satisfaction and use intention. As such, cashierless shopping systems present digital innovations that are positively perceived by shoppers and may give stationary retailing a modern feel to rival online grocery shopping.

Based on the empirical results, retailers have to empower shoppers regarding cashierless systems. Shoppers want to have control over the checkout process and receive direct information, most relevantly in the mobile app scenario when shoppers can just walk out of the store. This is similar to customers’ perception of being in control of their personal data (Pantano et al., 2023). Consequently, customer convenience is almost always related shoppers’ perception of privacy issues. All our cashierless scenarios confirm previous findings in this regard (Ponte and Bonazzi, 2021; Yu et al., 2024). Shoppers perceive data privacy as a challenge for these systems, which relates to the risks regarding the loss of control over their personal data. The effect of privacy concerns was most pronounced in the face recognition process in which shoppers rely on biometrical identification. This is not unexpected, because this type of cashierless process and its reliance on biometrical identification is the most unknown type of check-out system. Lesser familiarity entails rising concerns of data privacy and leads to increasing barriers for retailers to implementing face recognition processes (de Bellis and Johar, 2020). This is in line with Gao et al. (2023) who show that use experience mitigates a social presence effect.

Customers also expect cashierless shopping systems to perform as expected, as generally supported in extant research (Lee and Pan, 2023; Moriuchi, 2021; Ponte and Bonazzi, 2021). By performance expectancy, customers reflect on how useful and helpful the cashierless systems are. Cashierless shopping systems must offer a valuable service that enhances the experience, such as eliminating waiting time, thus freeing up time for other activities. Whereas performance expectancy determines both satisfaction and behavioral intention, effort expectancy is only relevant for shoppers’ willingness to use cashierless shopping systems. Shoppers expect the systems to be easy to use with minimal effort and believe they would quickly learn how to operate the cashierless systems. While effort expectancy is generally supported in previous research (Lee and Pan, 2023; Zhang and Kang, 2019), the present findings add more details and insights into its effect.

The empirical results yield no effect of participants’ technology affinity on either self-service satisfaction or intention to use cashierless systems. This is in contrast to the relevance of technophobia (Subero-Navarro et al., 2021) but confirms the findings regarding technology anxiety with app checkout systems (Chuawatcharin and Gerdsri, 2019). These results raise the question of whether cashierless systems may present a touchpoint of innovation in stationary retailing. Even though the technology behind these systems incorporates the latest artificial intelligence, shoppers may not attribute these systems to digitization and innovation.

Whether cashierless shopping systems reduce costs highly depends on the comparison of costs for staffing the checkout and maintaining the technology. However, introducing more technology potentially answers the need to find staff in the first place. Nonetheless, recent literature shows that even autonomous stores, i.e., stores without on-site staff, are faced with staffing challenges. For example, customers ask for on-site staff to ensure that questions and issues can be directly solved. Customers seem to be attracted by staffed stores per se. Benoit et al. (2024), for instance, highlight that staffed stores outperform autonomous stores and suggest that autonomous stores should be located in stand-alone settings to avoid competing with staffed stores.

Following the empirical results and the discussion, we next explicitly answer our research questions (see Fig. 1).

Firstly, research question 1 targets the general willingness to adopt cashierless shopping systems. Consumers are generally willing to adopt and use cashierless shopping systems. Regardless of the specific system (e.g., mobile app or facial recognition), performance expectancy and social influence positively affect shoppers’ behavioral intention. In contrast, perceived data privacy has a negative impact. Notably, consumers’ effort expectancy influences the intention to use mobile apps for the checkout process.

Secondly, research question 2 looks at customer benefits and how they contribute to consumers’ self-service satisfaction and adoption of cashierless shopping systems. Customer benefits that contribute to satisfaction and the use of cashierless systems include empowerment, experience, and performance expectancy. These factors enhance both customer satisfaction and the willingness to use the innovative systems. The results also confirm the negative influence of perceived data privacy

on self-service satisfaction and behavioral intention to use the cashierless systems.

Lastly, research question 3 explores customer-relevant factors to extend our understanding of cashierless shopping systems. Convenience is a significant factor that drives the usage of cashierless systems. However, social, security, and technical concerns hinder their adoption. Additionally, consumers' use of cashierless systems depends on the specific shopping situation.

8. Theoretical and managerial implications

8.1. Theoretical implications

Following our research agenda, the present study provides numerous theoretical implications that collectively contribute to the theoretical framework and the acceptance of digital innovations in stationary retailing.

The customer benefit adoption model extends existing technology acceptance theories by integrating key consumer-centric value dimensions (i.e., convenience, empowerment, and experience) into the UTAUT. While UTAUT primarily focuses on cognitive and social influences on technology adoption, our model incorporates experiential and psychological benefits that shape consumer perceptions in retail contexts. This addition enhances theoretical understanding by demonstrating that adoption decisions are not solely rational but also influenced by affective and situational factors. By shifting the focus from purely functional drivers to customer-centered experiences, the model challenges the assumption that utilitarian factors alone drive technology acceptance. Instead, it highlights the importance of perceived control and engagement in shaping consumer behavior. Furthermore, by incorporating data privacy concerns as a critical barrier, the model accounts for the increasing importance of trust and perceived risk in digital interactions, contributing to a broader understanding of consumer adoption beyond structured retail settings and offering insights applicable to other self-service innovations such as autonomous stores, AI-driven customer interfaces, and biometric payment systems. By bridging the gap between technology adoption and consumer experience literature, the model provides a more holistic framework that advances research on self-service technologies and enhances the applicability of digital retail innovations. The framework can organize future research on emerging retail technologies.

The initial study validates the established UTAUT (Venkatesh et al., 2003) for customer acceptance of innovative technologies in retail. The empirical studies highlight the factors driving customers' willingness to adopt cashierless systems. Across both studies, we consolidate the importance of predominantly rational factors, namely performance and effort expectancy. The relevance of social influence toward the system was only supported in the first study. The qualitative insights indicate that, although social value may not be the primary focus regarding the cashierless system, social considerations are still prominent. Customers expressed concerns about social responsibility, social interactions within the store, and potential job losses.

The consumer benefit technology adoption model integrates key customer benefits of digitization in retail, i.e., customer convenience, empowerment, experience, and relevance. Whereas relevance is closely releated to the rational and functional uses of the cashierless systems mentioned, performance and effort expectancy, convenience, empowerment, and experience explicitly substantiate our understanding of such digital innovations. The integration further helps to develop a more comprehensive model that explains how these benefits impact the relationships between antecedents and customers' intention to use cashierless shopping systems. Based on the empirical results, customer benefits may exert both direct and indirect influences on customers' use intentions, though not in a uniform manner. Consequently, future research should consider the complex ways in which customer benefits influence preferences for digital technologies in retailing.

Last but not least, the qualitative results go beyond customer acceptance and customer benefits. The qualitative study explores additional topics that influence customer willingness or resistance to engage with cashierless shopping systems. Five key topics emerge: convenience, security concerns, social concerns, technical concerns, and shopping situation. These findings suggest that the theoretical focus on direct effects regarding customer characteristics only captures part of the complexity. Theory should place greater emphasis on customer characteristics and situational conditions, as they play a crucial role in shaping customer experience and acceptance over time.

8.2. Managerial implications

Retailers have to ensure intuitive and efficient system processes by correspondingly designing the user interface at the service encounter and ascertaining the interoperability with back-end systems. For innovative digital technologies at the checkout, e.g., app authentication and facial recognition, in stationary retailing, even minor technical issues can lead to customer dissatisfaction and rejection of cashierless shopping systems. Customers are particularly concerned with the system identifying articles wrongly, determining a wrong article quantity, and charging the wrong prices. People are also concerned that the system might charge their account multiple times. Data privacy concerns must also be addressed. Retailers need to reassure customers that their data and automated payment processes are secure. Beyond informational approaches, retailers can achieve this, for example, by minimizing processing times between self-checkout and the digital transmission of purchase receipts.

As agency shifts from cashiers to customers, the burden of proof falls to the customer, intensifying shopping stress and leading to the rejection of cashierless shopping systems. Retail and goverment policies have to address this agency shift by reorganizing the evidence process.

Security concerns need to be further accounted for by guidelines for system security standards and data protection laws. While these standards should not be prohibitive against any cashierless shopping system, they should put the customer in control.

Retailers should additionally empower customers to use cashierless shopping systems. A key procedural aspect is to provide customer control. Customers do not only have to be informed but also need to perceive that they have a choice. One option is that cashierless systems are not used instead but in combination with traditional processes. Such implementation is similar to human-robot teams. Customer confidence can also be strengthened by a human-in-the-loop.

Financial incentives and communicational campaigns can also motivate customers to use these systems. Retailers can, for example, provide discounts for using cashierless systems. Communication should emphasize convenience benefits, such as reduced waiting times, and mitigate social concerns. Such campaigns should aim to enhance customers' self-esteem and social acceptance. Influencers may be positive role models to further encourage the use of cashierless shopping systems.

Lastly, the adoption of cashierless shopping systems introduces new competitive dynamics between small and large retailers, potentially widening the gap in technological capabilities and customer expectations. Large retailers have the financial resources and infrastructure to implement advanced self-service technologies at scale. They can leverage automation to enhance efficiency, reduce operational costs, and provide seamless shopping experiences. These investments can strengthen customer satisfaction and loyalty by offering convenience-driven differentiation, making it increasingly difficult for smaller retailers to compete on service and technological innovation. However, small retailers can counterbalance this advantage by focusing on personalized customer service, niche offerings, and community engagement, elements that large retailers may struggle to replicate in automated environments. Nonetheless, we envision hybrid cashier areas that combine cashierless shopping systems with human interaction.

Such a portfolio cares for diverse consumer preferences, particularly among customers who value social engagement and personalized assistance. Policymakers and industry stakeholders should consider support mechanisms, such as technology-sharing initiatives, so that small retailers are not left behind in an increasingly digitalized retail landscape.

9. Limitations and future research

The multi-step research process sheds light on cashierless shopping systems in food retailing. While the research methodology aligns well with the study's objectives, certain limitations remain and provide opportunities for further research.

First, the initial adoption of cashierless shopping systems poses a significant challenge to customer acceptance, necessitating strategic implementation by companies with supportive measures. Although the research considers multiple points in time, the analyses rely on cross-sectional data. However, user experience is crucial in shaping customer satisfaction with these systems and develops over time. Thus, longitudinal observations can provide further insights into cashierless system acceptance and usage. Research can also extend to customer characteristics and situational conditions to improve our understanding further.

Second, while our samples align with the research objectives and reflect diversity in age, gender, and income, they are focused on German shoppers in grocery retailing. Future studies can broaden the cultural context, for example toward higher power distance, collectivism, and lower uncertainty avoidance. Further research can also incorporate additional characteristics, such as social norms, values, beliefs, and time orientation. This would enhance the generalizability of the findings.

Third, our studies rely on the well-established UTAUT, into which we integrate customer convenience, empowerment, and experience as well as self-service satisfaction. We also control for demographic variables but do not consider psychographic variables, e.g., customer's personality, in this context (e.g., Paetz and Schultz, 2025). However, the inclusion of such variables may provide further insights into the boundary conditions when using cashierless systems. We encourage further research to consider such boundary conditions and additional control variables in the presented benefit-based customer adoption model.

Fourth, our qualitative exploration identifies five relevant topics that can further contribute to the understanding of cashierless shopping systems. The customer voices provide additional information on convenience, security concerns, social concerns, technical concerns, and shopping situations. All areas warrant future research that can build on the exploration presented. The discussion on customer benefits may be considerably affected by the shopping situation and driven by customer values regarding ethical, moral, and social dimensions. The qualitative findings suggest that when agency shifts in processes (e.g, customer is the cashier or customer needs to justify basket discrepancies), the perspective becomes a pain point for the customer. Future research may explore these shifts and provide managerial recommendations to mitigate the potential loss of customer benefits before becoming a pain point.

Lastly, our study concentrates on one type of cashierless shopping system per participant in the study approach. The studies also broadly capture each checkout system. Future research can delve deeper into specific situations and system contexts for each cashierless shopping system, particularly digitally innovative approaches. Instead of an either-or decision, we envision retailers embracing a combination of cashier and cashierless systems. Such a combination can mitigate some customer concerns but creates considerable managerial challenges that need further academic exploration.

CRediT authorship contribution statement

Carsten D. Schultz: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Friederike Paetz:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization.

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Declaration of competing interest

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The first author declares a guest editorship in the Journal of Retailing and Consumer Services. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX. Measurement Items of Study 1 and Study 2

Item	Statement
Performance Expectancy (Venkatesh et al., 2003; Im et al., 2011; Zhou, 2012)	
PE1	I would find the [cashierless system] useful in my shopping.
PE2	Using the [cashierless system] enables me to accomplish tasks independently and more quickly.
PE3	By using the [cashierless system], I save time so that I can focus on other things in my everyday life.
PE4	The [cashierless system] offers me added value while shopping.
PE5	Using the [cashierless system] increases my productivity.
Effort Expectancy (Venkatesh et al., 2003; Im et al., 2011; Zhou, 2012)	
EE1	The use of the [cashierless system] is clear and easy to understand.
EE2	It would be easy for me to become skillful at using the [cashierless system].
EE3	I would find the [cashierless system] easy to use.
EE4	Learning to use the [cashierless system] is easy for me.
Social Influence (Venkatesh et al., 2003; Walsh et al., 2013)	
	The use of the [cashierless system] …
SI1	… would help me feel accepted.
SI2	… would improve the way I am perceived.
SI3	… would make a good impression on other people.
SI4	… would give its owner social approval.
Technology Affinity (Franke et al., 2019)	
TA1	I like to occupy myself in greater detail with technical systems.
TA2	I like testing the functions of new technical systems.

(*continued on next page*)

(*continued*)

Item	Statement
TA3	I predominantly deal with technical systems because I have to.
TA4	When I have a new technical system in front of me, I try it out intensively.
TA5	I enjoy spending time becoming acquainted with a new technical system.
TA6	It is enough for me that a technical system works; I don't care how or why.
TA7	I try to understand how a technical system exactly works.
TA8	It is enough for me to know the basic functions of a technical system.
TA9	I try to make full use of the capabilities of a technical system.
Data Privacy (Zhang et al., 2019)	
DP1	I am concerned that the [cashierless system] will collect too much personal information from me.
DP2	I am concerned that the [cashierless system] will use my personal information for other purposes without my authorization.
DP3	I am concerned that the [cashierless system] will share my personal information with other entities without my authorization.
Behavioral Intention to Use Cashierless Systems (Brooke, 1996)	
BI1	I think that I would like to use the [cashierless system] frequently.
BI2	I found the [cashierless system] unnecessarily complex.
BI3	I think that I would need the support of a technical person to be able to use the [cashierless system].
BI4	I think that the various functions of the [cashierless system] can be integrated well into retail shopping.
BI5	I think that the [cashierless system] is not fully mature yet.
BI6	I can imagine that most people would learn to use the [cashierless system] very quickly.
BI7	I found the [cashierless system] very cumbersome to use.
BI8	I felt very confident using the [cashierless system].
BI9	I needed to learn a lot of before I could use the [cashierless system].

APPENDIX. Measurement Items continued for Study 2

Item	Statement
Customer Convenience (Duarte et al., 2022)	
CC1	I need a lot of time to use the [cashierless system].
CC2	The payment process using the [cashierless system] is very fast.
CC3	The waiting time for using the [cashierless system] is very short.
CC4	I can save time by using the [cashierless system].
CC5	The [cashierless system] makes it easy for me to complete my shopping.
CC6	It doesn't take long to pay for my purchase at the [cashierless system].
Customer Empowerment (Hunter and Garnefeld, 2008; Zhu et al., 2007)	
CE1	In my dealings with this [cashierless system], I feel I am in control.
CE2	The ability to use the [cashierless system] is advantageous for me.
CE3	I feel good because I have the choice to use the [cashierless system].
CE4	Using the [cashierless system] gives me more options than before.
CE5	I have the choice to use the [cashierless system] if I want to.
Customer Experience (Gong et al., 2021)	
CX1	I am passionate about using the [cashierless system].
CX2	I would love to use the [cashierless system].
CX3	Using the [cashierless system] would get my attention.
CX4	I would become immersed in the use of the [cashierless system].
Social Anxiety (Fenigstein et al., 1975)	
SA1	It takes me time to overcome my shyness in new situations.
SA2	I have trouble working when someone is watching me.
SA3	I get embarrassed very easily.
SA4	I find it hard to talk to strangers.
SA5	I feel anxious when I speak in front of a group.
SA6	Large groups make me nervous.
Self-service Satisfaction (Sharma et al., 2021)	
SS1	dissatisfied … satisfied
SS2	upset … content
SS3	unhappy … happy
SS4	displeased … pleased

APPENDIX. Overview of Sample Characteristics for Study 1

Table
Overview of Sample Characteristics

Variable		Absolute Frequency	Relative Frequency
Gender	female	229	56.4%
	male	177	43.6%
	diverse	–	–
Age	≤19	9	2.2%
	20–29	201	49.5%

(*continued on next page*)

Table (*continued*)

Variable		Absolute Frequency	Relative Frequency
	30–39	99	24.4%
	40–49	33	8.1%
	50–59	39	9.6%
	60-69+	22	5.4%
	≥70	3	0.7%
Net Household Income	<1000	40	9.9%
	1000–1999	68	16.7%
	2000–2999	128	31.5%
	3000–3999	64	15.8%
	4000–4999	43	10.6%
	≥5000	46	11.3%
	w/o answer	17	4.2%
Experience	yes	129	31.8%
	no	277	68.2%

Appendix. Group Comparisons

Table
Group Comparison Gender

Independent	Dependent	female	male	*p*-value
Performance Expectancy	Self-service Satisfaction	0.275	0.118	0.098
Effort Expectancy	Self-service Satisfaction	−0.009	−0.014	0.453
Social Influence	Self-service Satisfaction	−0.024	−0.010	0.371
Technology Affinity	Self-service Satisfaction	−0.054	−0.041	0.395
Social Anxiety	Self-service Satisfaction	0.038	0.002	0.208
Data Privacy	Self-service Satisfaction	−0.113	−0.047	0.142
Customer Convenience	Self-service Satisfaction	−0.034	0.010	0.325
Customer Empowerment	Self-service Satisfaction	0.190	0.302	0.136
Customer Experience	Self-service Satisfaction	0.407	0.421	0.436
Performance Expectancy	Behavioral Intention	0.105	0.119	0.442
Effort Expectancy	Behavioral Intention	0.162	0.189	0.334
Social Influence	Behavioral Intention	−0.021	−0.042	0.316
Technology Affinity	Behavioral Intention	−0.036	0.011	0.187
Social Anxiety	Behavioral Intention	0.013	−0.023	0.345
Data Privacy	Behavioral Intention	−0.200	−0.167	0.316
Customer Convenience	Behavioral Intention	0.040	0.062	0.365
Customer Empowerment	Behavioral Intention	0.178	0.289	0.107
Customer Experience	Behavioral Intention	0.175	0.189	0.431
Self-service Satisfaction	Behavioral Intention	0.243	0.131	0.062

Table
Group Comparison Age

Independent	Dependent	≥45	<45	*p*-value
Performance Expectancy	Self-service Satisfaction	0.197	0.146	0.341
Effort Expectancy	Self-service Satisfaction	0.005	−0.008	0.435
Social Influence	Self-service Satisfaction	−0.014	−0.017	0.493
Technology Affinity	Self-service Satisfaction	−0.044	−0.064	0.340
Social Anxiety	Self-service Satisfaction	0.011	0.026	0.379
Data Privacy	Self-service Satisfaction	−0.055	−0.160	0.058
Customer Convenience	Self-service Satisfaction	0.022	−0.026	0.306
Customer Empowerment	Self-service Satisfaction	0.224	0.310	0.220
Customer Experience	Self-service Satisfaction	0.422	0.366	0.313
Performance Expectancy	Behavioral Intention	0.206	0.061	0.062
Effort Expectancy	Behavioral Intention	0.152	0.199	0.192
Social Influence	Behavioral Intention	−0.008	−0.042	0.195
Technology Affinity	Behavioral Intention	0.025	−0.029	0.129
Social Anxiety	Behavioral Intention	0.010	0.019	0.373
Data Privacy	Behavioral Intention	−0.176	−0.224	0.187
Customer Convenience	Behavioral Intention	−0.003	0.099	0.064
Customer Empowerment	Behavioral Intention	0.260	0.108	0.113
Customer Experience	Behavioral Intention	0.195	0.159	0.318
Self-service Satisfaction	Behavioral Intention	0.189	0.172	0.414

Appendix. Total Effects on Behavioral Intention

Independent	Dependent	Effect (95% CI)	LLCI	ULCI
Performance Expectancy	Behavioral Intention	0.166	0.069	0.267
Effort Expectancy	Behavioral Intention	0.159	0.099	0.219
Social Influence	Behavioral Intention	−0.036	−0.073	0.002
Technology Affinity	Behavioral Intention	−0.016	−0.063	0.036
Social Anxiety	Behavioral Intention	0.007	−0.045	0.053
Data Privacy	Behavioral Intention	−0.212	−0.269	−0.157
Customer Convenience	Behavioral Intention	0.047	−0.021	0.116
Customer Empowerment	Behavioral Intention	0.265	0.178	0.350
Customer Experience	Behavioral Intention	0.256	0.176	0.331
Self-service Satisfaction	Behavioral Intention	0.180	0.110	0.250

Notes: LLCI: lower level confidence interval, ULCI: upper level confidence interval.

Data availability

Data will be made available on request.

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